

ADVANCE DIGITAL IMAGE PROCESSING LECTURE # 2

DIGITAL IMAGE FUNDAMENTALS

Topics to Cover

- ❑ Image acquisition using Array sensor
- ❑ Cross section of human eye
- ❑ Sampling and Quantization

- ❑ Spatial and Gray level Resolution

- ❑ Image Scaling
 - ✓ Image Zooming
 - ✓ Image Shrinking

- ❑ Basic relationship between pixels
 - ✓ Neighbors
 - ✓ Adjacency
 - ✓ Path
 - ✓ Connectivity
 - ✓ Regions and Boundaries
 - ✓ Distance between pixels

Image Acquisition using Array Sensor

- ❑ Figure shows individual sensors arranged in a form of 2-D array
- ❑ This arrangement exists in modern day digital cameras
- ❑ A typical sensors for these cameras is a Charged Coupled Device (C C D) array, which can be manufactured with a broad range of sensing properties and can be packaged in arrays of 4000 x 4000 elements or more
- ❑ CCD sensors are used widely in digital cameras and other light sensing instruments
- ❑ The response of each sensor is proportional to the integral of the light energy projected on to the surface of the sensor

Array sensor

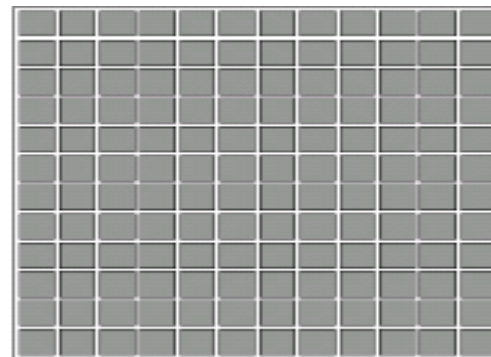
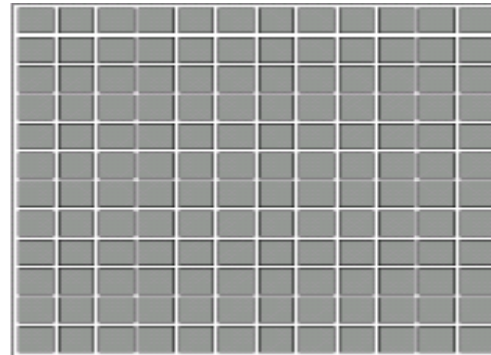


Image Acquisition using Array Sensor

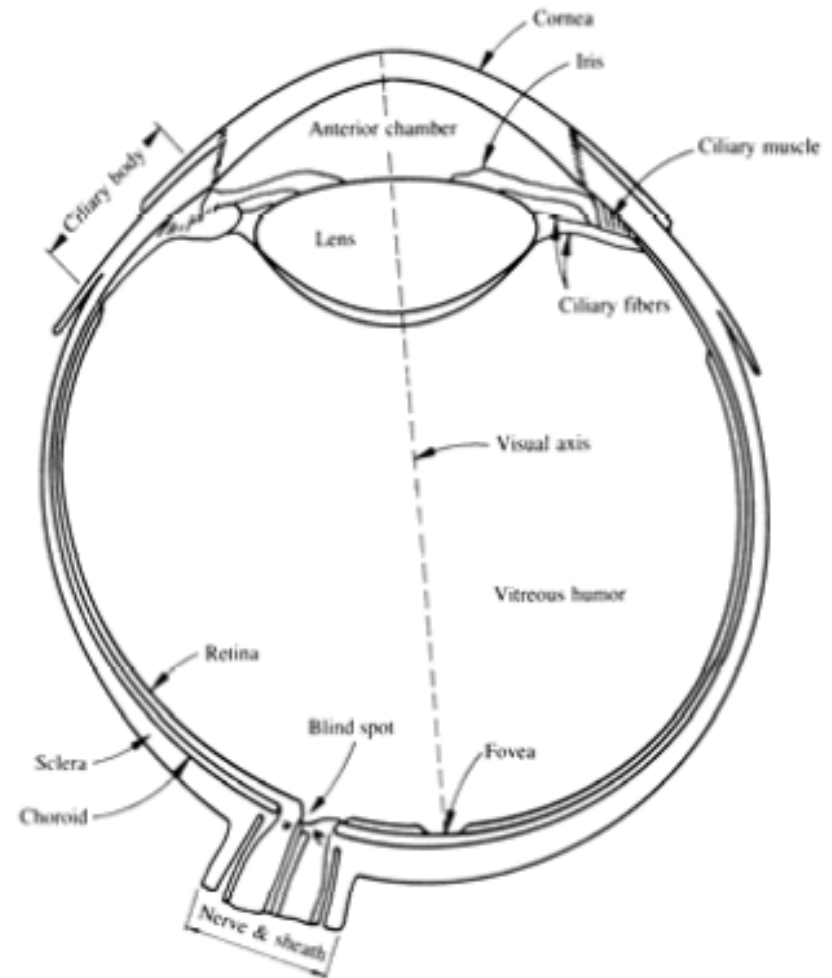
- ❑ Sense the light reflection on the sensor (arranged in 2D form)
- ❑ Voltage waveform will be received (Analog signal)
- ❑ Convert this analog signal into digital signal through sampling and quantization
 - ✓ Apply Sampling to digitize coordinate values
 - ✓ Apply Quantization to digitize amplitude values
- ❑ Store the digitized value in memory

Array sensor



Cross section of the human eye

- Eye characteristics
 - nearly spherical
 - approximately 20 mm in diameter
 - three membranes
 - cornea (transparent) & sclera (opaque) outer cover
 - choroid contains a network of blood vessels, heavily pigmented to reduce amount of extraneous light entering the eye. Also contains the iris diaphragm (2-8 mm to allow variable amount of light into the eye)
 - retina is the inner most membrane, objects are imaged on the surface



Retinal surface

- Retinal surface is covered in discrete light receptors
- Two classes
 - Cones
 - 6-7 million located primarily near the center of the retina (the *fovea*)
 - highly sensitive to color
 - can resolve fine details because each is attached to a single nerve ending
 - Cone vision is called *photopic* or bright-light vision
 - Rods
 - 75-150 million distributed over the retinal surface
 - multiple rods connected to a single nerve ending
 - give a general overall picture of the field of illumination
 - not color sensitive but are sensitive to low levels of illumination
 - Rod vision is called *scotopic* or dim-light vision

Imaging in the eye

- Variable thickness lens: thick for close focus, thin for distant focus
- Distance of focal center of the lens to the retina (14-17 mm)
- Image of a 15m tree at 100m
 $15/100 = X/17$ or approximately 2.55 mm
- Image is almost entirely on the fovea

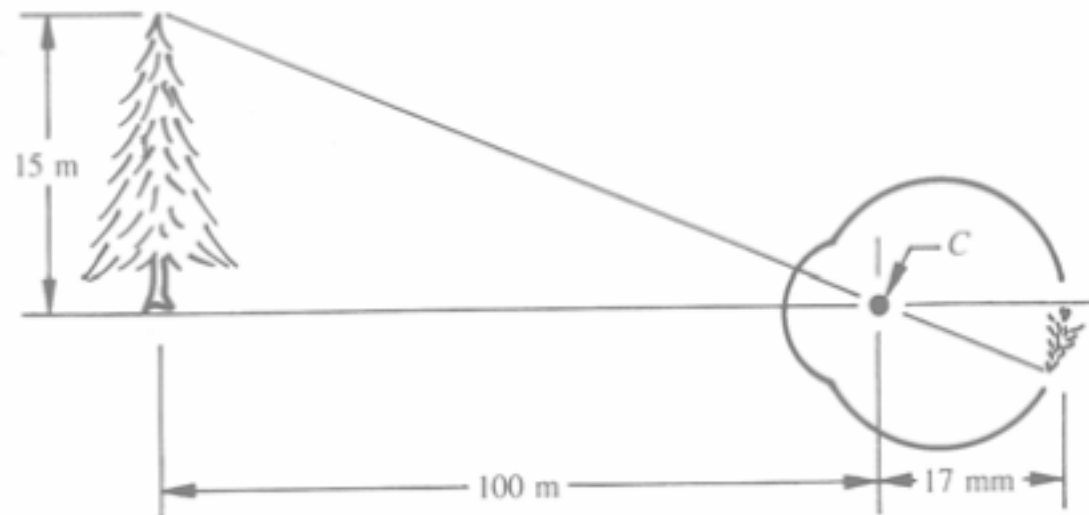


Image Digitization

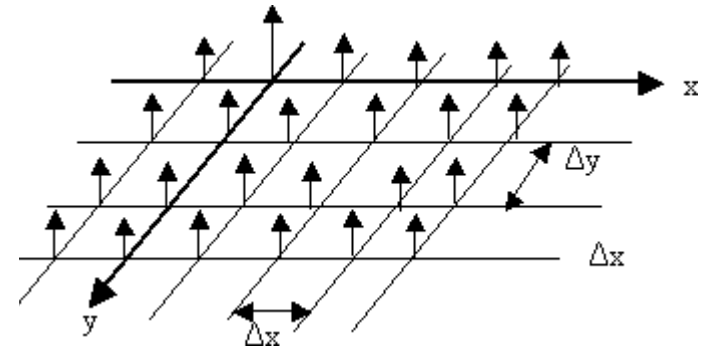
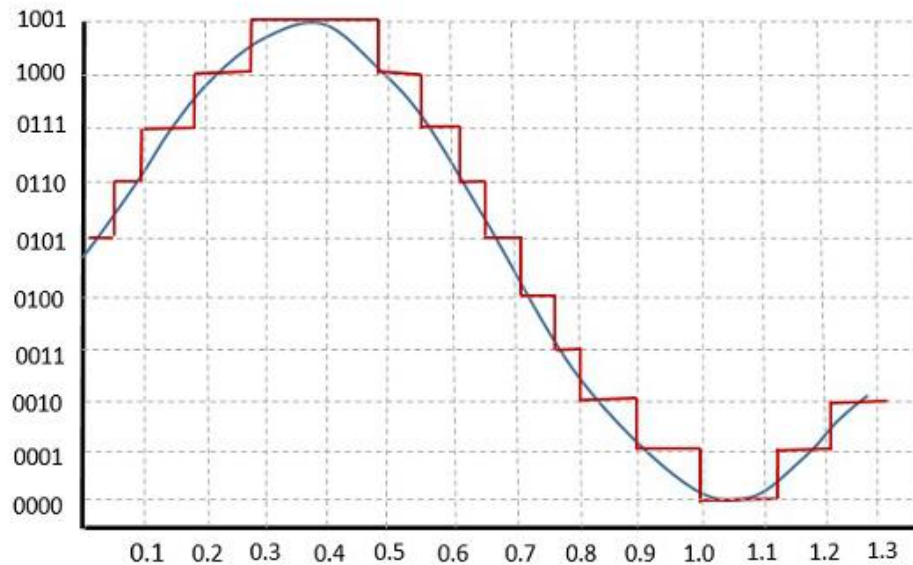
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- Digitalization of an analog signal involves two operations:
 - ✓ Sampling, and
 - ✓ Quantization.
- Both operations correspond to a discretization of a quantity, but in different domains.
- An image captured by a sensor is expressed as a continuous function $f(x,y)$ of two co-ordinates in the plane.
- Image digitization means that the function $f(x,y)$ is sampled into a matrix with M rows and N columns.
- The image quantization assigns to each continuous sample an integer value. The continuous range of the image function $f(x,y)$ is split into k intervals.
- The finer the sampling (i.e., the larger M and N) and quantization (the larger K) the better the approximation of the continuous image function $f(x,y)$.

Sampling

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- A continuous image function $f(x,y)$ can be sampled using a discrete grid of sampling points in the plane.
- Two neighboring sampling points are separated by distance Δx along the x axis and Δy along the y axis. Distances Δx and Δy are called the **sampling interval** and the matrix of samples constitutes the discrete image

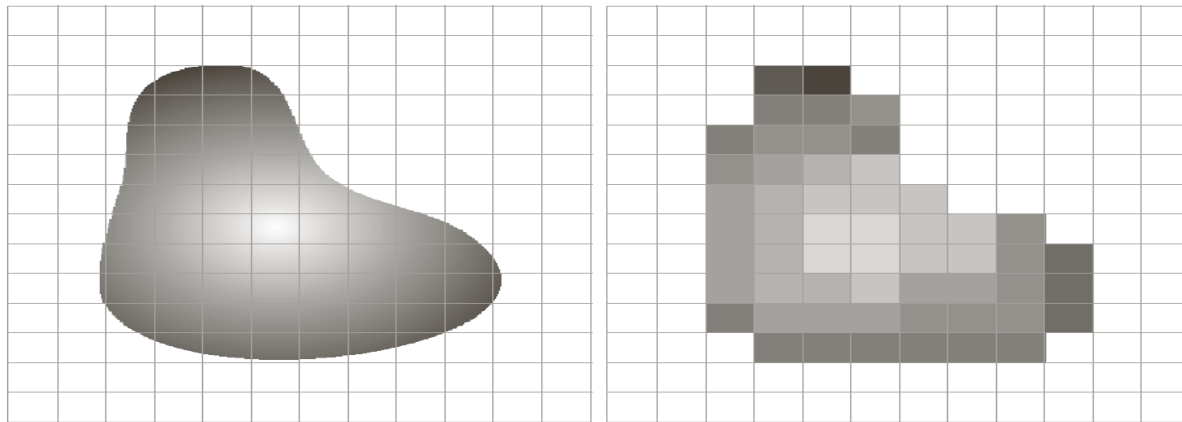


Sampling

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- Sampling corresponds to a discretization of the space. That is, of the domain of the function, into $f : [1; :: : ;N] \quad [1; :: : ;M]$
- Thus, the image can be seen as matrix,

$$f = \begin{bmatrix} f(1,1) & f(1,2) & \cdots & f(1,M) \\ f(2,1) & f(2,2) & \cdots & f(2,M) \\ \vdots & \vdots & \ddots & \vdots \\ f(N,1) & f(N,2) & \cdots & f(N,M) \end{bmatrix}$$



a b

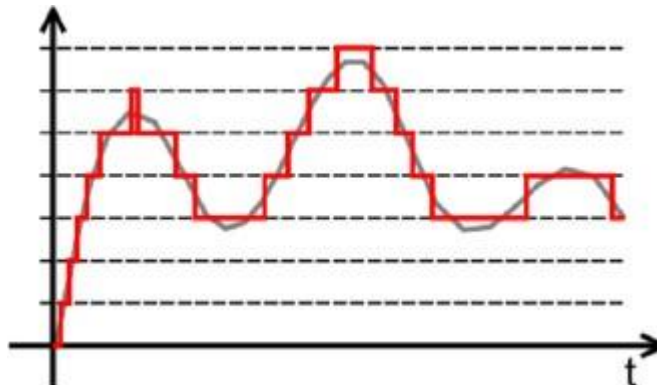
FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

Quantization

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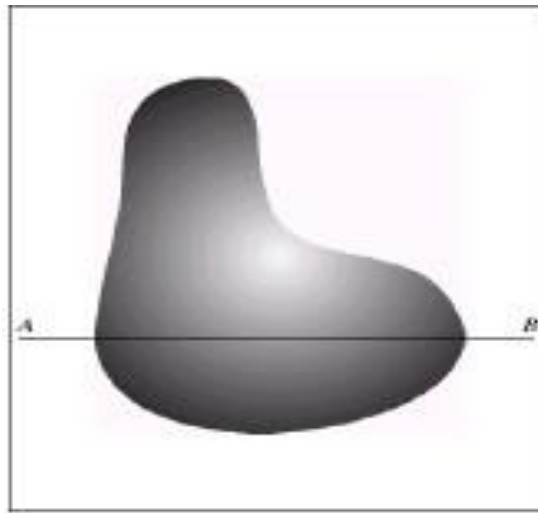
- The transition between continuous values of the image function (brightness) and its digital equivalent is called quantization.
- Quantization corresponds to a discretization of the intensity values.
- The number of quantization levels should be high enough for human perception of fine shading details in the image.
- After Sampling and Quantization we get the

following $f : [1; :: : ;N] [1; :: : ;M] \rightarrow [0,, L]$

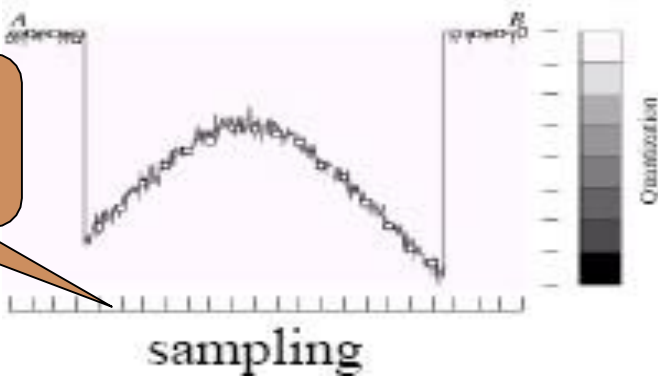


Sampling and Quantization

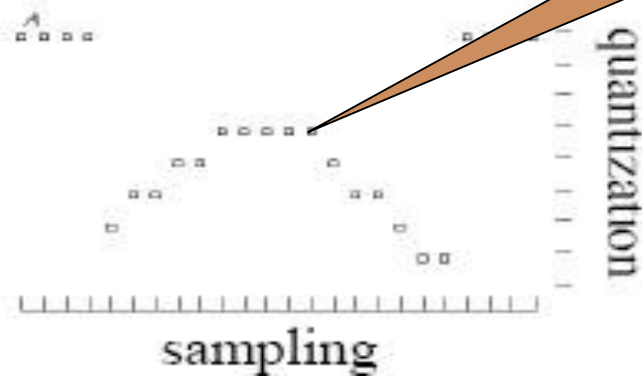
8



Digitizing the
coordinate
values



Digitizing the
amplitude values



Sampling and Quantization

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- ❑ **Sampling:** Digitization of the spatial coordinates (x,y)
- ❑ **Quantization:** Digitization in amplitude (also called *gray level quantization*)
- ❑ **8 bit quantization:** $2^8=256$ gray levels (0: black, 255: white)
- ❑ **Binary (1 bit quantization):** 2 gray levels (0: black, 1: white)
- ❑ **Commonly used number of samples (resolution)**
- ❑ Digital still cameras: 640x480, 1024x1024, up to 4064 x 2704
- ❑ Digital video cameras: 640x480 at 30 frames/second and higher

Spatial and Gray level Resolution

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□ Spatial resolution

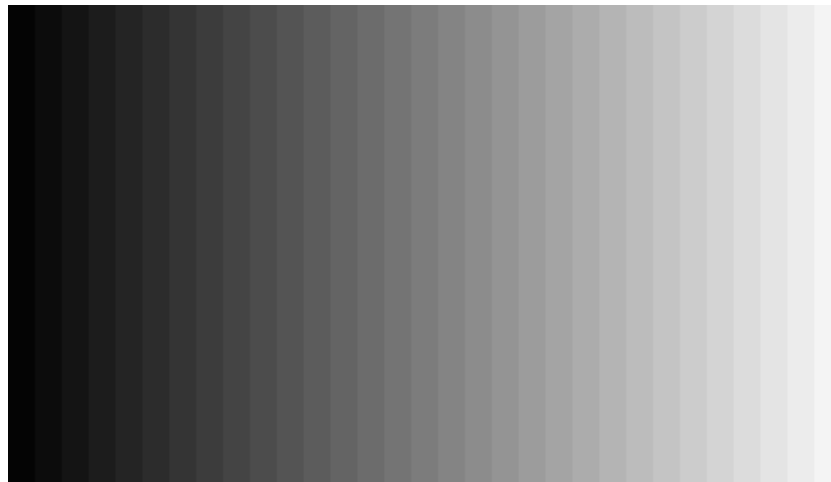
Resolution in Space

- ✓ A measure of the smallest visible detail in an image stated with *dots (pixels) per unit distance, dots per inch (dpi)*
- ✓ *No. of pixels specifies the spatial resolution*

□ Intensity or Gray level resolution

Resolution in Gray values

- ✓ The smallest visible change in intensity or gray level stated with *8 bits, 16 bits, etc.*



Spatial and Gray level Resolution

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- The Digitization process requires to determine M, N and L



- M and N are Spatial Resolution
- $L = \text{Gray level Resolution}$, k is total number of bits
 - $L = 2^k$, where L represents Gray level
- Image Storage = Spatial Resolution * Gray level Resolution
- The no. of bits required to store the image is::
 - $b = M \times N \times k$ or $b = N^2 \times k$
- Sampling --→ Spatial Resolution
- Quantization --→ Gray level Resolution

Spatial and Gray level Resolution

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TABLE 2.1

Number of storage bits for various values of N and k .

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

Spatial and Gray level Resolution

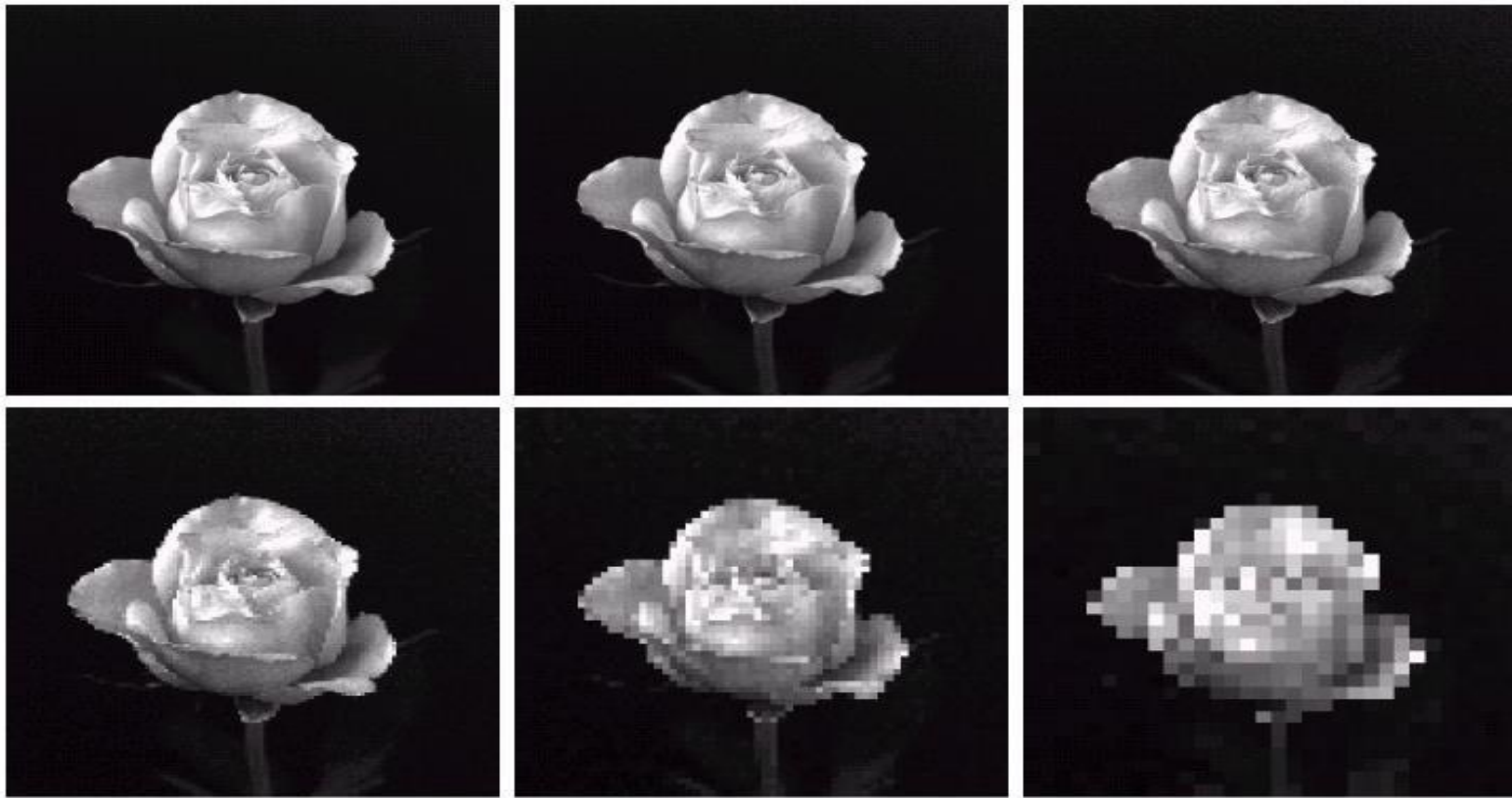
13



Variation in Spatial Resolution from 1024 x 1024 to 32 x 32

Spatial and Gray level Resolution

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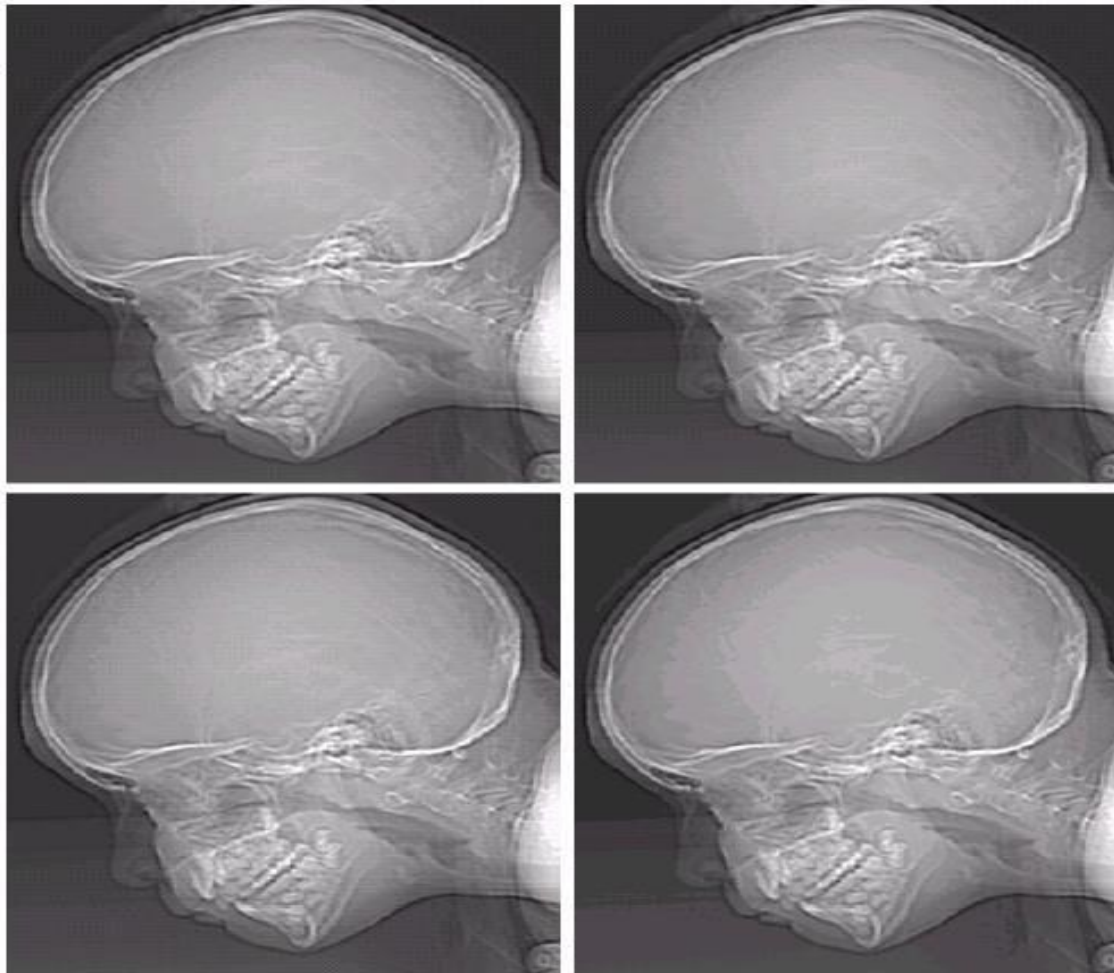


a b c
d e f

FIGURE 2.20 (a) 1024×1024 , 8-bit image. (b) 512×512 image resampled into 1024×1024 pixels by row and column duplication. (c) through (f) 256×256 , 128×128 , 64×64 , and 32×32 images resampled into 1024×1024 pixels.

Spatial and Gray level Resolution

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a b
c d

FIGURE 2.21

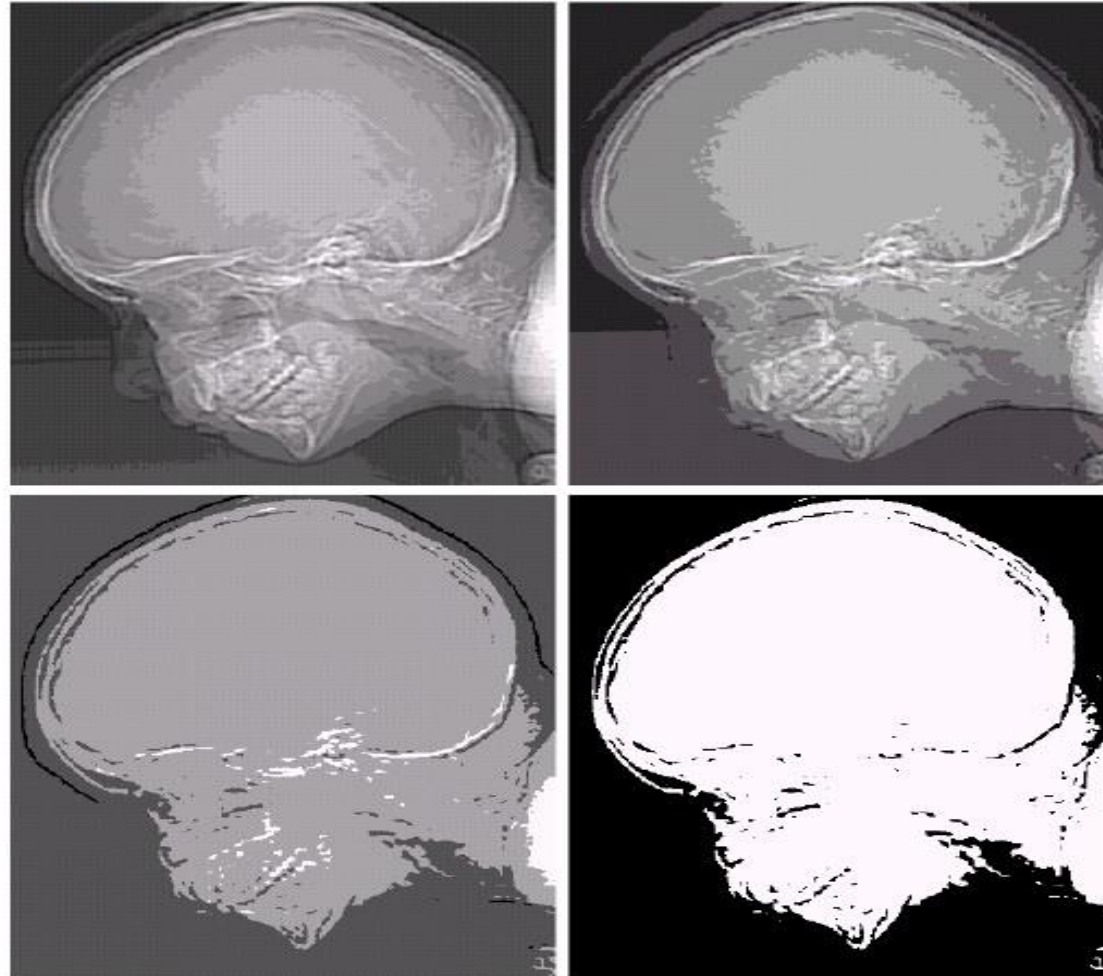
(a) 452×374 , 256-level image. (b)–(d) Image displayed in 128, 64, and 32 gray levels, while keeping the spatial resolution constant.

Spatial and Gray level Resolution

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e f
g h

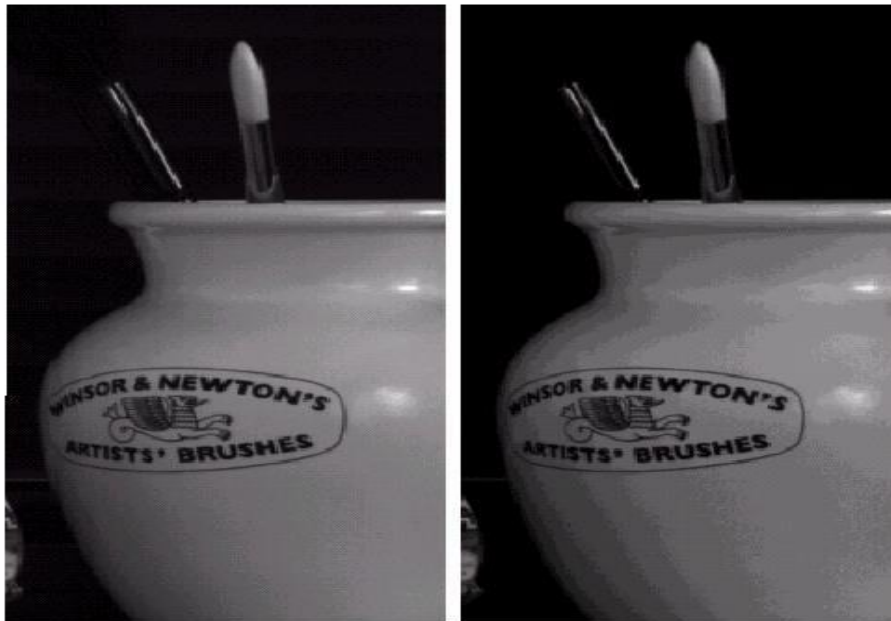
FIGURE 2.21
(Continued)
(e)–(h) Image
displayed in 16, 8,
4, and 2 gray
levels. (Original
courtesy of
Dr. David
R. Pickens,
Department of
Radiology &
Radiological
Sciences,
Vanderbilt
University
Medical Center.)



Spatial and Gray level Resolution

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- ❑ The representation of an image with insufficient number of gray levels produces false edges or boundaries in an image, a phenomenon known as False Contouring or Contouring defect
- ❑ False Contouring is quite visible in images displayed using 16 or less gray levels as shown in the images of the skeleton in the previous slide



Contouring
defect

Spatial and Gray level Resolution

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Medium level of detail

Low level of detail

Large amount of detail



a b c

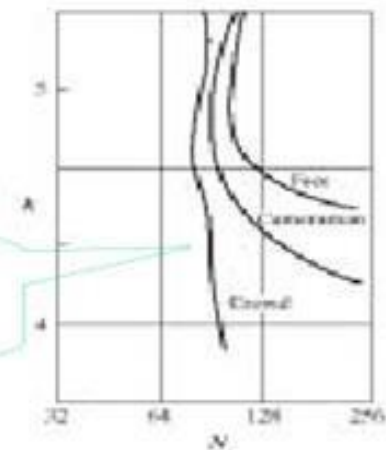
FIGURE 2.22 (a) Image with a low level of detail. (b) Image with a medium level of detail. (c) Image with a relatively large amount of detail. (Image (b) courtesy of the Massachusetts Institute of Technology.)

Spatial and Gray level Resolution

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- Varying N and k , the observers are then asked to rank the image according to their subjective quality.
- The results are summarized in the form of so-called **isopreference curve**.
- The **isopreference curve** tends to become more vertical as the detail of the image increases.

FIGURE 2.25
Representative
isopreference
curves for the
three types of
images in
Fig. 2.22.



Nearly vertical $\Rightarrow$ for a given N , the perceived quality for this type of image is nearly independent of the number of gray levels

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IMAGE SCALING

Simple scaling



Image Scaling

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- ❑ Zooming (up scaling, resizing upward) requires two steps
 - ✓ The Creation of new pixel locations
 - ✓ Assignment of gray levels to new pixel locations

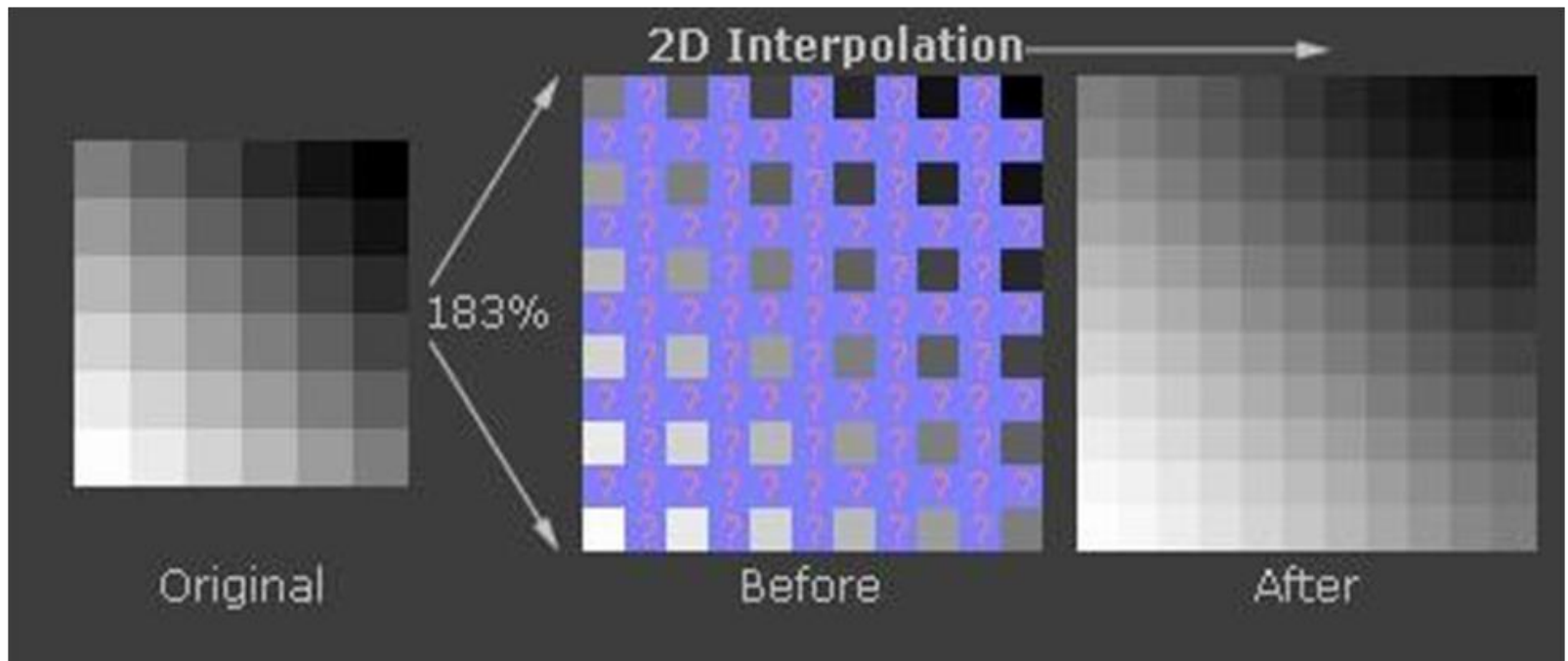


Image Scaling via Interpolation

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- ❑ Common interpolation algorithms can be grouped into two categories:
 - ✓ Adaptive and
 - ✓ Non-adaptive.

Adaptive vs Non-Adaptive Algorithm

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- ❑ **Adaptive algorithms** has the capability of applying different level of interpolation on different area on an image
- ❑ Adaptive methods change depending on what they are interpolating (sharp edges vs. smooth texture)
- ❑ It include many algorithms in licensed software such as: Qimage, PhotoZoom Pro, Genuine Fractals and others.
- ❑ **Non-adaptive algorithms** treat all pixels equally, they include: bilinear, bicubic, pixel replication etc..
- ❑ The more adjacent pixels they include, the more accurate they can become, but this comes at the expense of much longer processing time.

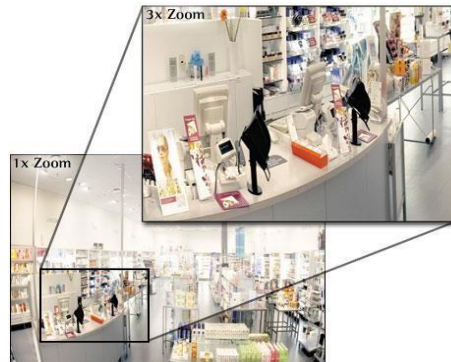


Image Scaling

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Pixel Replication

- ✓ Pixel replication is applicable when we want to increase the size of an image an integer number of times.
- ✓ For example to double the size of an image we can duplicate each column, this doubles the size of image in horizontal direction. Then we duplicate each row of the enlarged image to double the size in the vertical direction
- ✓ The same procedure can be applied to enlarge the image by any integer number of times (triple, quadruple and so on)
- ✓ The gray level assignment of each pixel is predetermined by the fact that new locations are exact duplicate of old locations

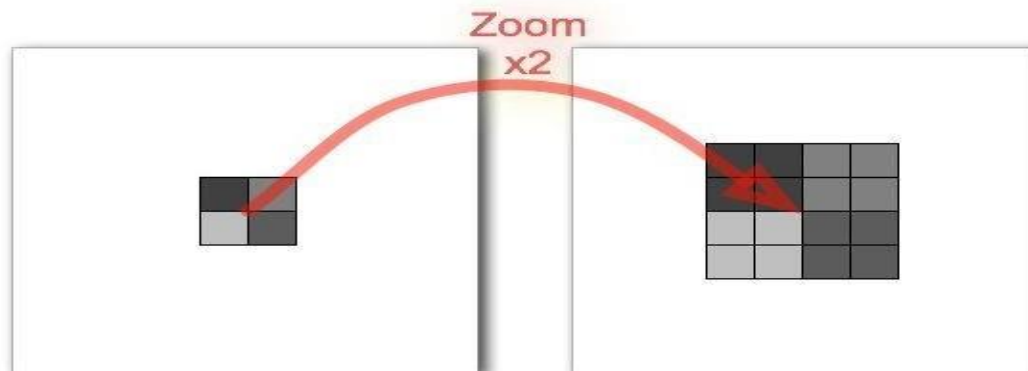


Image Scaling

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□ Bilinear Interpolation [2]

- In the mathematical field of numerical analysis, **interpolation** is a method of constructing new data points within the range of a discrete set of known data points.
- Bilinear interpolation considers the closest 2x2 neighborhood of known pixel values surrounding the unknown pixel.
- It then takes a weighted average of these 4 pixels to arrive at its final interpolated value. This results in much smoother looking images than pixel replication technique.
- The diagram below is for a case when all known pixel distances are equal, so the interpolated value is simply their sum divided by four.
- In case the distance varies then The closer pixels are given more weightage in the calculation

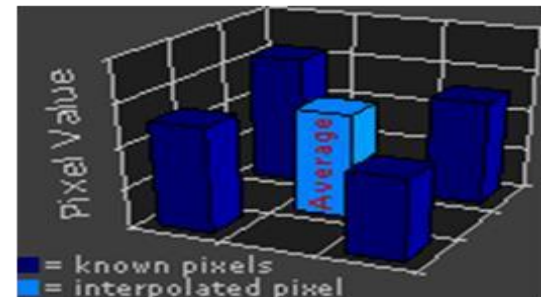


Image Scaling

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- The key idea is to perform linear interpolation first in one direction, and then again in the other direction.
- Suppose that we want to find the value of the unknown function f at the point $P = (x, y)$.
- It is assumed that we know the value of f at the four points $Q_{11} = (x_1, y_1)$, $Q_{12} = (x_1, y_2)$, $Q_{21} = (x_2, y_1)$, and $Q_{22} = (x_2, y_2)$.

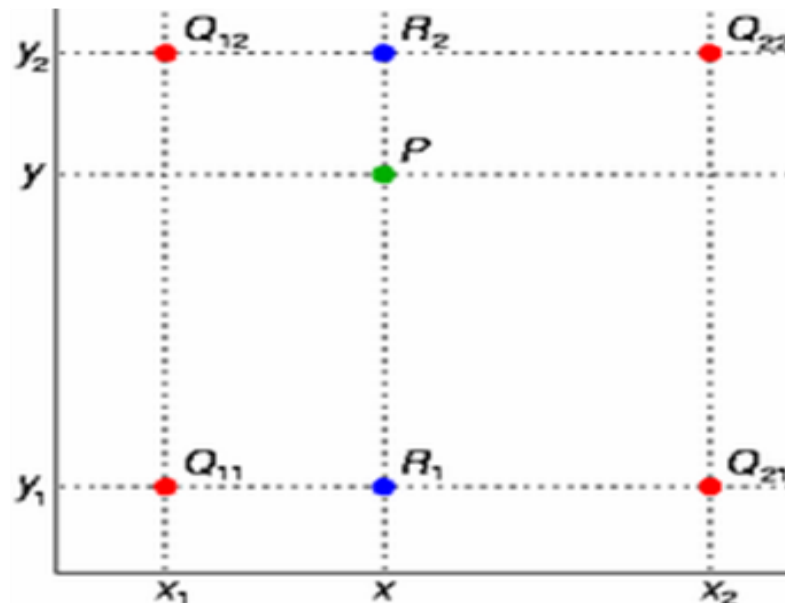


Image Scaling

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We first do linear interpolation in the x-direction. This yields

$$f(R_1) \approx \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21})$$

where $R_1 = (x, y_1)$,

$$f(R_2) \approx \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22})$$

where $R_2 = (x, y_2)$.

$$f(P) \approx \frac{y_2 - y}{y_2 - y_1} f(R_1) + \frac{y - y_1}{y_2 - y_1} f(R_2).$$

We proceed by interpolating in the y-direction.

Image Scaling

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This gives us the desired estimate of $f(x, y)$.

$$\begin{aligned} f(x, y) \approx & \frac{f(Q_{11})}{(x_2 - x_1)(y_2 - y_1)}(x_2 - x)(y_2 - y) \\ & + \frac{f(Q_{21})}{(x_2 - x_1)(y_2 - y_1)}(x - x_1)(y_2 - y) \\ & + \frac{f(Q_{12})}{(x_2 - x_1)(y_2 - y_1)}(x_2 - x)(y - y_1) \\ & + \frac{f(Q_{22})}{(x_2 - x_1)(y_2 - y_1)}(x - x_1)(y - y_1). \end{aligned}$$

Image Scaling

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❑ Bicubic Interpolation [3]

- ✓ Bicubic goes one step beyond bilinear by considering the closest 4x4 neighborhood of known pixels-- for a total of 16 pixels.
- ✓ Since these are at various distances from the unknown pixel, closer pixels are given a higher weighting in the calculation.
- ✓ Bicubic produces noticeably sharper images than the previous two methods, and is perhaps the ideal combination of processing time and output quality
- ✓ For this reason it is a standard in many image editing programs (including Adobe Photoshop), printer drivers and in-camera interpolation

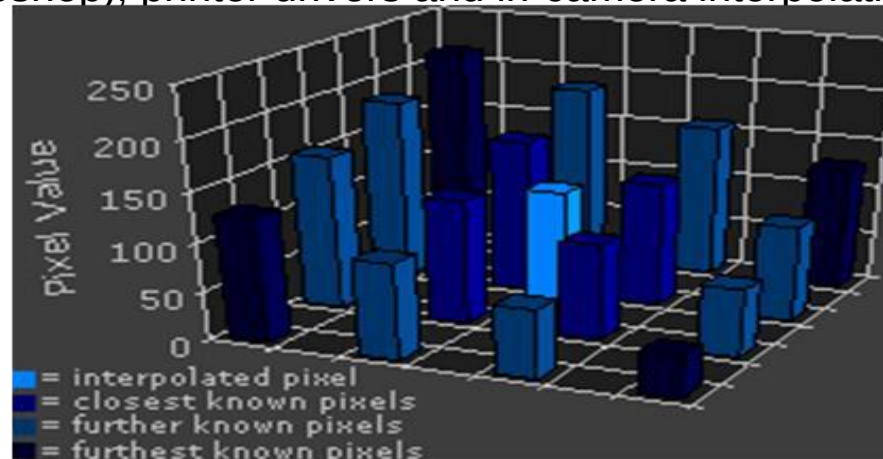
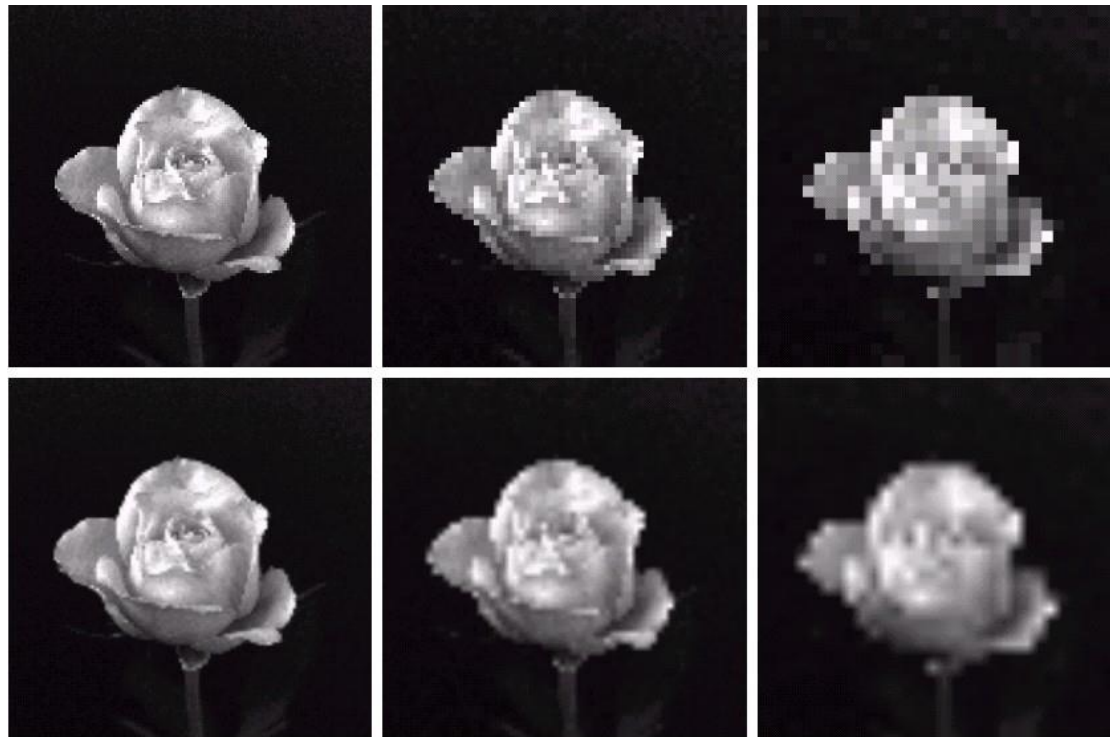


Image Scaling

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Increasing Spatial Resolution



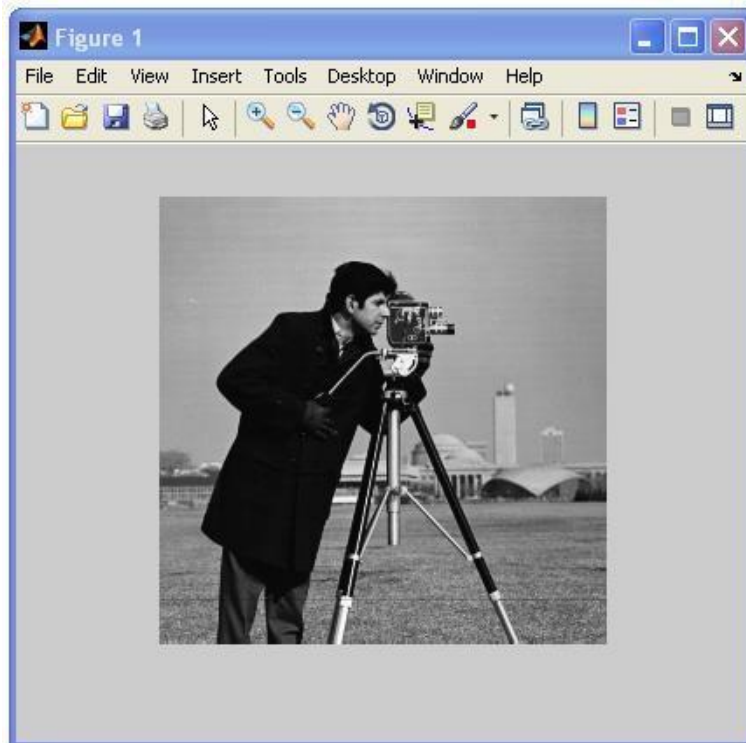
a	b	c
d	e	f

FIGURE 2.25 Top row: images zoomed from 128×128 , 64×64 , and 32×32 pixels to 1024×1024 pixels, using nearest neighbor gray-level interpolation. Bottom row: same sequence, but using bilinear interpolation.

Image Scaling

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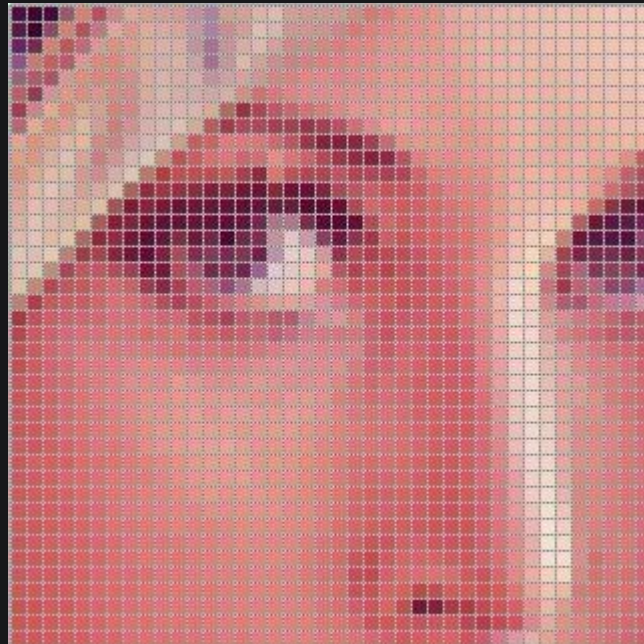
- ❑ Shrinking (Down scaling, resizing downward)
 - ✓ Image shrinking is done in the similar manner as zooming with one difference as now the process of pixel replication is row column deletion. Now we can delete every second column and row for shrinking



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Basic Relationship between pixels



Basic Relationship between Pixels

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- ❑ Neighborhood
- ❑ Adjacency
- ❑ Paths
- ❑ Connectivity
- ❑ Regions and boundaries

0	1	0	1	1

1	—	1	—	1
0	1	0	1	1

0	1	—	1	0
0	0	1	0	0



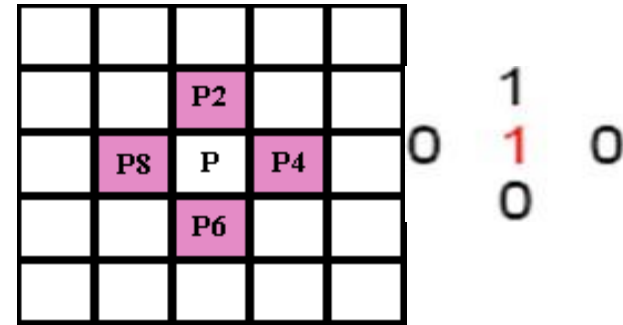
Pixels Neighborhood

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Neighbors of a pixel p at coordinates (x,y)

✓ 4-neighbors of p , denoted by $N_4(p)$:

$(x-1, y)$, $(x+1, y)$, $(x, y-1)$, and $(x, y+1)$



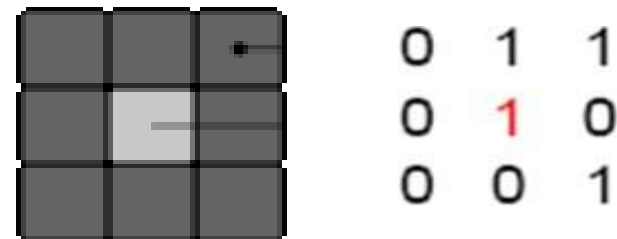
✓ 4 diagonal neighbors of p , denoted by $N_D(p)$:

$(x-1, y-1)$, $(x+1, y+1)$, $(x+1, y-1)$, and $(x-1, y+1)$



✓ 8 neighbors of p , denoted $N_8(p)$

$N_8(p) = N_4(p) \cup N_D(p)$



Adjacency

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- ❑ Two pixels are said to be adjacent in if
 - ✓ They are neighbors
 - ✓ Their intensity level V are the same [binary or gray levels are equal]
 - For example in a binary image with values 0 and 1, two pixels may be 4-neighbors, but they are said to be connected only if they have the same value
- ❑ Let V be the set of gray level values used to define adjacency
- ❑ In a binary image, $V=\{1\}$ if we are referring to adjacency of pixels with value 1
- ❑ In gray scale image the idea is same but set V contains more elements. For example, in the adjacency of pixels with the range of possible gray level values 0 to 255, set V could be any subset of these 256 values

Adjacency

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- Let V be the set of intensity values
- 4-adjacency:** Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$
e.g. $V = \{0, 1\}$

1	1	0
1	1	0
1	0	1

- p in **RED** color
- q can be any value in **GREEN** color.

Adjacency

- **8-adjacency:** Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$

e.g. $V = \{1, 2\}$

0	1	1
0	2	0
0	0	1

- p in **RED** color
- q can be any value in **GREEN** color

Adjacency

□ **m-adjacency:** Two pixels p and q with values from V are m-adjacent if

(i) q is in the set $N_4(p)$, or

(ii) q is in the set $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V

$q \in N_D(p)$ and $N_4(p) \cap N_4(q) = \emptyset$

e.g. $V = \{ 1 \}$

- b & c are m-adjacent
- b & e are m-adjacent
- e & i are m-adjacent
- e & c are NOT m-adjacent

0 _a	1 _b	1 _c
0 _d	1 _e	0 _f
0 _g	0 _h	1 _i

Path

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- A (digital) path (or curve) from pixel p with coordinates (x, y) to pixel q with coordinates (s, t) is a sequence of distinct pixels with coordinates (x_0, y_0) , (x_1, y_1) , $\dots$, (x_n, y_n)

Where $(x_0, y_0) = (x, y)$, $(x_n, y_n) = (s, t)$

(x_i, y_i) and (x_{i-1}, y_{i-1}) are adjacent for $1 \leq i \leq n$.

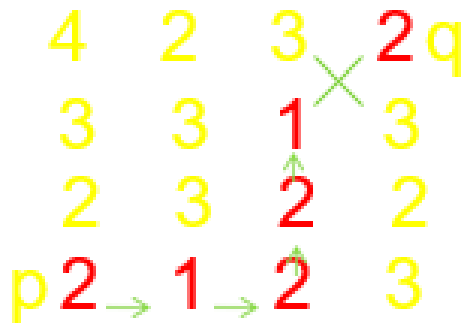
- Here n is the *length* of the path.
- If $(x_0, y_0) = (x_n, y_n)$, the path is *closed* path.
- We can define 4-, 8-, and m-paths based on the type of adjacency used.

Examples: Adjacency and Path

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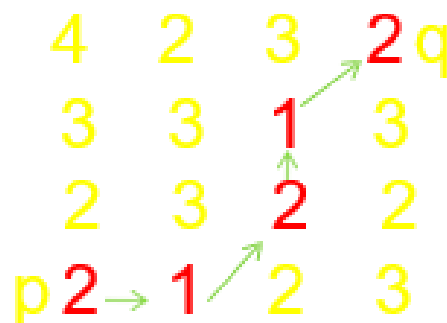
Consider the image segment shown in figure. Compute length of the **shortest-4**, **shortest-8** & **shortest-m** paths between pixels p & q where, $V = \{1, 2\}$.

Shortest 4



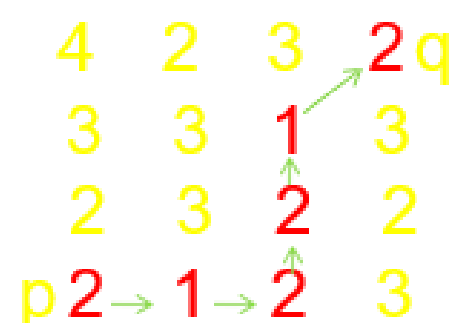
No path

shortest 8



path = 4

shortest m



path = 5

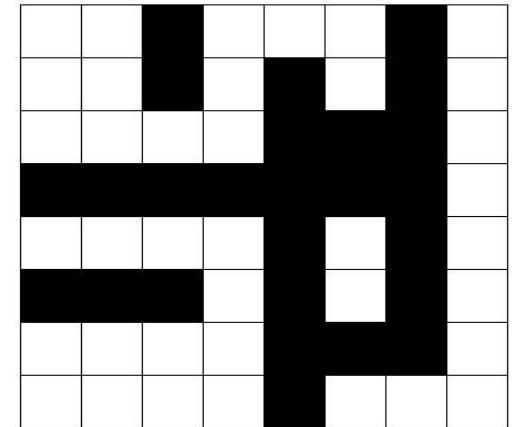
Connectivity

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- Let S represents a subset of pixels in an image. Two pixels p with coordinates (x_0, y_0) and q with coordinates (x_n, y_n) are said to be **connected in S** if there exists a path between them consisting entirely of pixels in S

e.g $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$

- Where $\forall i, 0 \leq i \leq n, (x_i, y_i) \in S$



- For any pixel p in S , the set of pixels that are connected to it in S is called a **connected component** of S
- If it only has one connected component, then set S is called **Connected set**
- Distinct connected components in S are said to be **disjoint**

Connected Component Labeling

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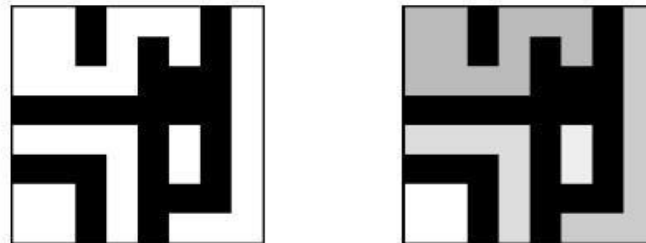
- Ability to assign different labels to various disjoint connected components of an image
- Connected component labeling is a fundamental step in automated image analysis (Shape, Area, Boundary)

1	1	0	1	1	1	0	1
1	1	0	1	0	1	0	1
1	1	1	1	0	0	0	1
0	0	0	0	0	0	0	1
1	1	1	1	0	1	0	1
0	0	0	1	0	1	0	1
1	1	0	1	0	0	0	1
1	1	0	1	0	1	1	1

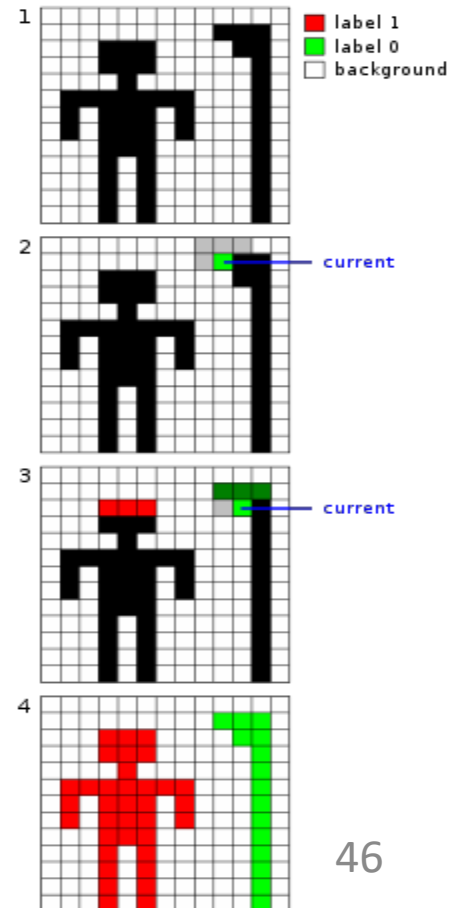
a) binary image

1	1	0	1	1	1	0	2
1	1	0	1	0	1	0	2
1	1	1	1	0	0	0	2
0	0	0	0	0	0	0	2
3	3	3	3	0	4	0	2
0	0	0	3	0	4	0	2
5	5	0	3	0	0	0	2
5	5	0	3	0	2	2	2

b) connected components labeling



c) binary image and labeling, expanded for viewing

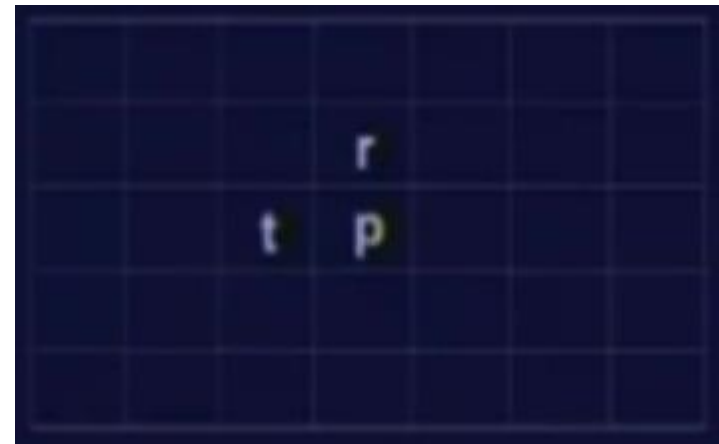
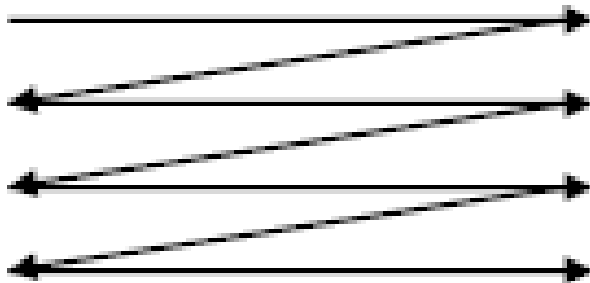


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Connected Component Labeling

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- ❑ Scan an image from left to right and from top to bottom.
- ❑ Assume 4-Connectivity
 - Let p denote the pixel of interest, and r and t denote the upper and left neighbors of p , respectively



- ❑ Before p , points r and t are scanned.

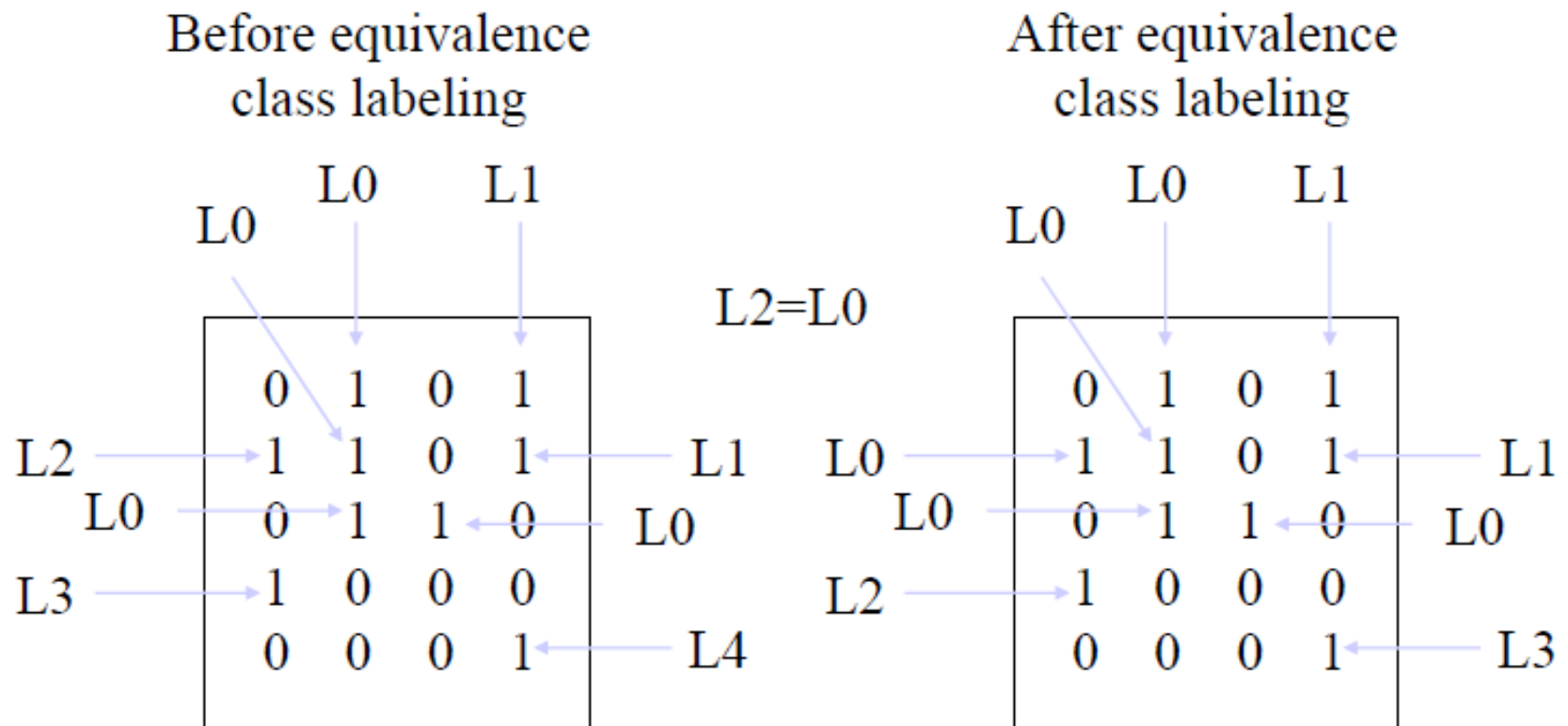
Connected Component Labeling

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- Consider the following procedure
 - if $p=0$ continue to the next position
 - if $r=t=0$ assign a new label to p (L_n)
 - if $r=t=1$ and they have the same label, assign that label to p
 - if only one of r and t are 1, assign its label to p
 - if $r=t=1$ and they have different labels, assign one label to p and note that the two labels are equivalent (that is r and t are connected through p)
- At the end of the scan, sort pairs of equivalent labels into equivalence classes and assign a different label to each class

Connected Component Labeling Example

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Regions & Boundaries

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- Let R represent a subset of pixels in an image
- We call R a region of the image if R is a connected set
- We can specify the region by using 4-adjacency and 8-adjacency
- Region = { Set of all pixels which fulfill adjacency criteria }
- Boundary (or border)
 - ✓ The *boundary* of the region R is the set of pixels in the region that have one or more neighbors that are not in R .
 - ✓ If R happens to be an entire image, then its boundary is defined as the set of pixels in the first and last rows and columns of the image.

- RED colored 1 is NOT a member of border if 4-connectivity is used between region and background. It is if 8-connectivity is used.

0	0	0	0	0
0	1	1	0	0
0	1	1	0	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

Distance measures

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- For pixels p , q and z with coordinates (x, y) , (s, t) , and (v, w) , respectively, D is a distance function or metric if
 - ✓ $D(p, q) \geq 0$ ($D(p, q) = 0$ if $p=q$)
 - ✓ $D(p, q) = D(q, p)$
 - ✓ $D(p, z) \leq (D(p, q) + D(q, z))$
- It is often useful in image processing to be able to calculate the distance between two pixels in an image, but this is not as straightforward as it seems.
- The presence of the pixel grid makes several so-called *distance metrics* possible which often give different answers to each other for the distance between the same pair of points. We consider the three most important ones.
 - ✓ Euclidean distance
 - ✓ City-block distance or D4 distance.
 - ✓ D8 distance or chessboard distance.

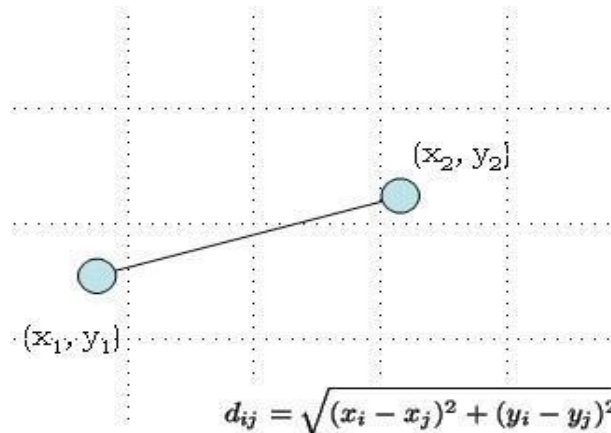
Distance measures

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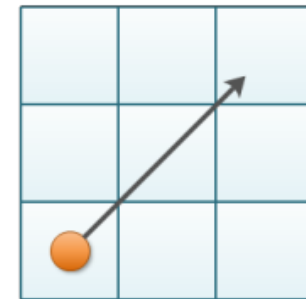
□ Euclidean distance

- ✓ If the two pixels p and q that we are considering have coordinates (x,y) and (s,t) , then the Euclidean distance between p and q is given by:

$$D_e(p, q) = [(x-s)^2 + (y-t)^2]^{1/2}$$



Euclidean Distance



$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

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Distance measures

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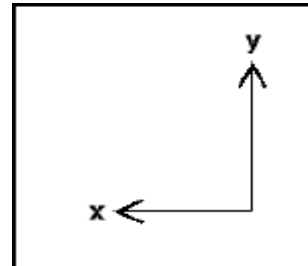
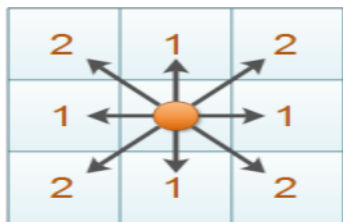
□ City-block distance or D4 distance or Manhattan distance

- ✓ This metric assumes that in going from one pixel to the other it is only possible to travel directly along pixel grid lines. Diagonal moves are not allowed.
- ✓ The distance between two points is the sum of the absolute differences of their coordinates
- ✓ The City-block distance between p and q is defined as

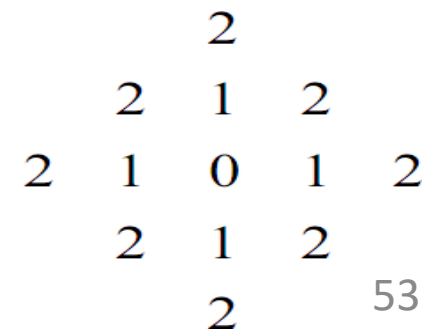
$$D4(p, q) = |x - s| + |y - t|$$

- ✓ The pixels having a distance D4 from (x, y) less than or equal to some value r form a diamond centered at (x, y). E.g the pixels with D4 distance ≤ 2 from (x, y) (the center point) form the following contours of constant distance

Manhattan Distance



Manhattan



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$$|x_1 - x_2| + |y_1 - y_2|$$

Distance measures

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□ D8 distance or chessboard distance or Chebyshev distance

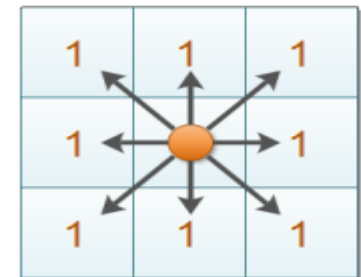
- ✓ This metric assumes that you can make moves on the pixel grid as if you were a King making moves in chess, *i.e.* a diagonal move counts the same as a horizontal move
- ✓ The distance between two pixels is the greatest of their differences of their coordinates

$$D8(p, q) = \max(|x - s|, |y - t|)$$

- ✓ The pixels having a D8 distance from (x, y) less than or equal to some value r from a square centered at (x, y) . E.g the pixels with D8 distance ≤ 2 from (x, y) (the center point) form the following contours of constant distance
- ✓ The pixels with $D8 = 1$ are the 8-neighbors of (x, y)

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Chebyshev Distance



$$\max(|x_1 - x_2|, |y_1 - y_2|)$$

References

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4. http://en.wikipedia.org/wiki/Connected-component_labeling
5. <http://www.mathworks.com/help/toolbox/images/f18-34186.html>
6. <http://homepages.inf.ed.ac.uk/rbf/HIPR2/distance.htm>

For any query Feel Free to ask



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